INCH-POUND

MIL-F-50945D (AR)
AMENDMENT 9
30 September 2002
SUPERSEDING
AMENDMENT 8
12 April 2001

MILITARY SPECIFICATION

FUZE, PD, M567/XM935 (M935)/M936 LESS BOOSTER

This amendment forms a part of MIL-F-50945D (AR), 08 September 1982, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "(M935)" after "M567/XM935"; and wherever it appears in the specification.

Title: Add "/M936" after "(M935)"; and where applicable.

PAGE 2

2.1.2:

Add to title of 9272193, "or Body Assembly, Front, M935".

Add to title of 9272194, "or Body Assembly, Front, M935".

Under DRAWINGS, Add the following:

"12972495 - Pellet".

PAGE 4

3.12.1: Delete in its entirety.

3.12.2: Delete in its entirety.

3.12.3: Delete in its entirety.

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PAGE 8

3.20, Add paragraph as follows:

"3.20 Lead assembly output. The lead assembly shall pierce a hole .220" minimum in diameter in a lead disc (dwg. 8816368)."

PAGE 10

4.3.1.2: Delete "Cap, Shipping... 9255255...10".

PAGE 18

Table I, First article inspection: delete "Cap, Shipping Dwg. 9255255, Examination for defects".

PAGE 20

Table I:

Under ballistic functioning at proving Ground: Add note (1) for Chg. 0, Set SQ and Chg. maximum, Set SQ.

Add note (1) as follows:

"(1) Samples for this test shall be separated into three groups; 30 samples temperature conditioned at plus 70 degrees F., 30 samples temperature conditioned at plus 145 degrees F., and 30 samples temperature conditioned at minus 50 degrees F. Temperature tolerance for conditioning shall be plus and minus 2.5 degrees F."

Under contractor's test facility: Delete "Transportation vibration; Ballistic functioning; Chg 0 set SQ; Chg 0 Set SQ, TV; Chg 0, Set Delay, TV."

Notes: Delete "d" and "e" in their entirety.

PAGE 21

Table I, Notes: Delete "k" and substitute:

"(k) Samples tested at the contractor's facility shall be assembled to the M567 configuration, dwg. 9246242 or the M935 configuration, dwg. 9255258, as applicable, unless otherwise determined by the contracting officer."

PAGE 23

4.4.2.1, Delete Major 112 in its entirety.

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4.4.2.6 and 4.4.2.6.1: Add Major 113 which reads as follows:

"113 Lubrication missing or improperly applied ...0.40% ...3.2 ...Visual".

PAGE 35

4.4.2.7:

* Delete Critical, 1 in its entirety and substitute the following:

"Critical I, 1 Spring pin missing 100% 3.2 4.5.18

Critical II, None defined"

Majors 101 and 102: Delete "0.40%" AQL and substitute "Note (a)" as follows:

"(a) Sample per paragraph 4.5.18.1".

PAGE 41

* 4.4.2.11: Delete Critical, I in its entirety and substitute the following:

"Critical I, 1 Groove pin missing 100% 3.2 Visual

Critical II, None defined"

PAGE 43

4.4.2.13, Add Major 106 which reads as follows:

"106 Lubrication missing or improperly applied ...0.40% ...3.2 ...Visual".

PAGE 44

4.4.2.14, Add Major 109 which reads as follows:

"109 Lubrication missing or improperly applied ...0.40% ...3.2...Visual".

PAGE 45

4.4.2.15, Delete Major 104 in its entirety.

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PAGE 54

* 4.4.2.22, Delete Critical, 1 in its entirety and substitute the following:

"Critical I, 1 Freedom of Latch Test
Firing pin not blocked by latch in upright position 100% 3.4 4.5.3

Critical II, None defined"

PAGE 55

* 4.4.2.22.1 Delete Criticals 1, 2, and 3 in their entirety and substitute the following:

"Critical I, 1 Slider Cavity Function and Safety Test
Arming pin does not block gage 100% 3.6 4.5.4 (a)

2 Firing pin does not block gage 100% 3.6 4.5.4 (a)

3 Critical Check Test
Firing pin or striker not blocked at specified heights 100% 3.8 4.5.7

Critical II, None defined"

PAGE 62

4.4.2.26, Add Major 105 which reads as follows:

"105 Lubrication missing or improperly applied ...0.40% ...3.2 ...Visual".

PAGE 63

* 4.4.2.27, Delete "Critical 1 in its entirety and substitute the following:

"Critical I, 1 Slider assembly not blocked by arming pin 100% 3.7 4.5.5 (a)

Critical II, None defined"

PAGE 72

4.4.2.35:

* Delete Critical 1 in its entirety and substitute the following:

"Critical II, A Length of safety pin 100% 3.2 Gage

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Critical I, None defined"

Delete Major 105 in its entirety and substitute the following:

"Major 105 Tensile test ...(b) ...3.1 ...Gage".

Major 106 Add new CD as follows:

"Major 106 Wrap test ...(b) ...3.1 ...Gage".

Note (b): Add at end of note the following:

"Test per MIL-STD-105D, inspection level S-4, AQL 1.0 percent for wrap test and level S-1, AQL 1.5 percent for the tension test.

Examination of wrap test samples shall utilize 4 X magnification."

PAGE 74

4.4.2.37:

* Delete Critical 1 and 2 in their entirety and substitute the followings:

"Critical I, 1 Pull wire not present or exceeds .27 inch above flush 100% 3.2 Gage

Critical I, 2 Detonator visible through lead hole (b) 100% 3.2 Gage

Critical II, A Pull wire not fully seated 100% 3.2 Gage"

Critical 2: Change inspection method from "visual" to "gage".

PAGE 82

4.4.2.41:

* Delete Critical 1 in its entirety and substitute the following:

"Critical I, 1 Pull wire not present or exceeds .27 inch above flush 100% 3.2 Gage

Critical II, A Pull wire not fully seated 100% 3.2 Gage "

* Delete Special A in its entirety ans substitute the following:

"Critical II, B Any or all of four critical inspection marks missing 100% 3.2 Visual"

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Critical 2: Change "Critical 2" to "Special A".

PAGE 83

4.4.2.42:

* Delete Criticals 1 and 2 in their entirety and substitute the following:

"Critical I, 1 Pull wire not present or exceeds .27 inch above flush 100% 3.2 Gage

Critical I, 2 Detonator visible through lead hole (b) 100% 3.2 Gage

Critical II, A Pull wire not fully seated 100% 3.2 Gage"

Critical 2: Change inspection method from "visual" to "gage".

Major 109: Add as follows:

"109 Presence of tape over rear face of body assembly ...0.40% ...3.2 ...Visual".

Major 106: Delete in its entirety.

PAGE 85

4.4.2.44: Delete in its entirety.

PAGE 86

4.4.2.45:

* Delete Critical 1 in its entirety and substitute the following:

"Critical I, 1 Pull wire not present or exceeds .27 inch above flush 100% 3.2 Gage

Critical II, A Pull wire not fully seated 100% 3.2 Gage "

* Delete Special, A in its entirety and substitute the following:

"Critical II, B Any or all of (4) critical inspection marks missing 100% 3.2 Visual"

Critical 2: Change "Critical 2" to "Special A".

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PAGE 87

4.4.2.46:

Contractor Test Facility :

* Delete Critical 1 in its entirety and substitute the following:

"Critical I, 1 Jolt and Jumble 12 (a) 3.10 4.5.10

Critical II, None defined"

Add Major 103 and Note (h) which reads as follows:

"103 Lubrication missing or improperly applied ...12 ...(h) ...3.2 ...Visual.

Same 12 samples after being used for the Jolt and Jumble Test shall be disassembled to examine for missing or improperly applied lubrication/grease (components to be checked are striker, setback pin, setback spring, slider detent, safety latch and selector cap O-ring). If one or more samples are classed defective, the lot shall be rejected."

Under contractor test facility:

Delete Criticals 2 and 3.

Delete Majors 101, 102, and 103.

Delete Notes b, c, d, e, f, g, and h.

PAGE 88

4.4.2.46:

Government Proving Ground

* Delete Criticals 1 and 2 in their entirety ans substitute the following:

"Critical I, 1 Premature functioning (a) 3.17.2 4.5.20

Critical I , 2 Arming safety 12 (b) 3.17.3 4.5.21

Critical II, None defined"

Majors 102 and Major 104: Delete "15" sample size and substitute "5".

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Under sample units: Delete "166" and substitute "146".

Majors 101 and 103, add Note (f) as follows:

"(f) Samples for this test shall be separated into three groups; 22 samples temperature conditioned at plus 70 degrees F., 20 samples temperature conditioned at plus 145 degrees F., and 20 samples temperature conditioned at minus 50 degrees F. Temperature tolerance for conditioning shall be plus and minus 2.5 degrees F."

PAGE 89

4.4.2.47:

Contractor Test Facility :

* Delete Critical 1 in it entirety ans substitute the following:

"Critical I, 1 Jolt and Jumble 12 (a) 3.10 4.5.10

Critical II, None defined"

Add Major 103 and Note (h) which reads as follows:

"103 Lubrication missing or improperly applied ...12 ...(h) ...3.2 ...Visual.

Note h. Same 12 samples after being used for the Jolt and Jumble Test shall be disassembled to examine for missing or improperly applied lubrication/grease (components to be checked are striker, setback pin, setback spring, slider detent, safety latch and selector cap O-ring). If one or more samples are classed defective, the lot shall be rejected."

Under contractor test facility:

Delete Criticals 2 and 3.

Delete Majors 101, 102, and 103.

Delete Notes b, c, d, e, f, g, and h.

PAGE 90

4.4.2.47:

Government Proving Ground :

* Delete Criticals 1 and 2 in their entirety and substitute the following:

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"Critical I, 1 Premature functioning (a) 3.17.2 4.5.20

Critical I , 2 Arming safety 12 (b) 3.17.3 4.5.21

Critical II, None defined"

Majors 102 and Major 104: Delete "15" sample size and substitute "5".

Under sampling units: Delete "166" and substitute "146".

Majors 101 and 103: Delete Note (f) and substitute:

"(f) Samples for this test shall be separated into three groups; 22 samples temperature conditioned at plus 70 degrees F., 20 samples temperature conditioned at plus 145 degrees F., and 20 samples temperature conditioned at minus 50 degrees F. Temperature tolerance for conditioning shall be plus and minus 2.5 degrees F."

PAGE 91

4.4.2.48, Add to title: "or Body Assembly, Front, M935".

PAGE 92

4.4.2.49, Add to title: "or Body Assembly, Front, M935".

PAGE 93

4.4.2.50, Delete Major 101 in its entirety.

4.4.2.51, Add new paragraph as included Page 93a

4.4.2.52, Add new paragraph as included Page 93b

PAGE 96

4.5.6, Change line 1 to read as follows:

"The assembly shall be hand-held or placed in".

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QUALITY CONFROMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.2.51	Cup Lead	SHEET 1 OF 1		9234684
CATEGORY	EXAMINATION or TEST	AQL OR 100 %	Requirements Paragraph	Nest Higher Assembly 9234683 Reference Paragraph Inspection method
Critical	None Defined			
Major				
101				
102	Thickness of base	0.40	3.2	Gage
103	Inside diameter	0.40	3.2	Gage
104	Outside diameter	0.40	3.2	Gage
105	Outside flange diameter	0.40	3.2	Gage
106	Thinckness of flange	0.40	3.2	Gage
107	Length from bottom to base of flange	0.40	3.2	Gage
	Corner radii (2 places)	0.40	3.2	Gage
Minor				
201	Protective Finish damaged	0.65	3.2	Visual
202	Surface finish improper	0.65	3.2	Visual
203	Evidence of poor workmanship	0.65	3.19	Visual
NOTES:				

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.52	TITLE Lead Assembly	SHEET 1 OF 1		DRAWING NUMBER 9234683
CATEGORY	EXAMINATION or TEST	AQL OR 100 %	Requirements Paragraph	Nest Higher Assembly 9255261 Reference Paragraph Inspection method
Critical	None Defined			
Major				
101	Overall length	0.40	3.2	Gage
102	Outside diameter	0.40	3.2	Gage
103	Charge depth improper	0.40	3.2	Gage
104	Lead assembly output test	0.40	3.20	4.5.22
Minor				
201	Evidence of poor workmanship	0.65	3.19	Visual
NOTES:				

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PAGE 97

4.5.9.1: Delete in its entirety.

PAGE 98

4.5.10:

Delete "the shipping cap (dwg. 9255255) shall be removed and".

Delete the first three sentences and substitute:

"The test shall be performed with M567 (dwg. 9246242) or the M935 (dwg. 9255258) as applicable."

PAGE 99

4.5.11.1: Delete in its entirety.

4.5.11.2: Add at end of paragraph as follows:

"Samples shall be temperature conditioned at the required temperatures for a minimum of six (6) hours and test fired within two minutes after removal from the conditioning chamber."

PAGE 100

4.5.18.1: Delete in its entirety and substitute the following:

"4.5.18.1 Examination for location and depth of spring pin cavity. The location and depth of the spring pin cavity shall be inspected in the following manner.

4.5.18.1.1 Initial production examination. Beginning with the first production shift, ten (10) consecutive drilled units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every two (2) hours until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the production represented by the units shall be rejected and the machine set-up shall be adjusted until ten (10) consecutive units are defect free.

4.5.18.2 Second production interval. After successfully completing the sampling of 4.5.18.1.1 for three consecutive eight hour production shifts with no defects, the sampling plan may be reduced to five (5) drilled units being inspected every four (4) hours and at the beginning and at the end of the shift. If a defect is found, the production represented by the sample shall be rejected and the sampling plan of 4.5.18.1.1 shall be reinstated.

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4.5.18.3 Third production interval. After successfully completing the sampling of 4.5.18.1.2 for five consecutive eight hour shifts, the sampling plan may be reduced to three (3) drilled units being inspected every four (4) hours and at the beginning and at the end of the shift. If a defect is found, the production represented by the sample shall be rejected and the sampling plan of 4.5.18.1.1 shall be reinstated.

4.5.18.4 Tooling change. A change of tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change. The above plan shall be applicable to each drilling machine equipment set up.

4.5.18.5 Production lapse. A stop in production of over one month will require starting the inspection at the initial production shift plan."

4.5.20.1:

Delete in its entirety and substitute:

"4.5.20.1 Functioning test. The sample fuzes shall be assembled to boosters and auxiliary boosters as appropriate. The fuzes shall then be assembled to the high explosive cartridge appropriate to the fuzes being tested (M374A3 for the M567 or M720 for the M935) and fired from the appropriate mortar between 45 and 75 degrees QE. When recovery of non-functioning fuzes is desired, the loaded sample fuzes (with boosters and auxiliary boosters as appropriate) shall be assembled to the appropriate inert material filled cartridges that simulate the weight and density of the normal HE fill and with an addition of a black powder pellet (12972495) to the rear of the boosters for a more distinctive function signature. If any fuze fails to cause the shell to function high order, that fuze shall be classified as defective (dud). If any fuze in either the SQ or delay setting mode causes the round to function prematurely, that fuze shall be classified as having a critical defect. The proving ground shall utilize a geophonic sensing device to determine delay functionings at maximum charge. Samples shall be temperature conditioned at the required temperatures for a minimum of six (6) hours and test fired within two (2) minutes after removal from the conditioning chamber."

Line 3: Delete "M374A2 and" Add at end of paragraph as follows: "Samples shall be temperature conditioned at the required temperatures for a minimum of six (6) hours and test fired within two minutes after removal from the conditioning chamber."

4.5.20.1.3, add at end of paragraph as follows:

"Groups of thirty-one shall be temperature conditioned as follows: eleven samples at plus 70 degrees F., ten samples @ plus 145 degrees F., and ten samples at minus 50 degrees F. Temperature tolerance for conditioning shall be plus and minus 2.5 degrees F. Samples shall be temperature conditioned at the required temperatures for a minimum of six (6) hours and test fired within two minutes after removal from the conditioning chamber."

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4.5.22 Add paragraph as follows:

"4.5.22 Test for deterioration of M53 delay element. One hundred (100) M53 delay shall be randomly selected from a lot of M53 delay element, which were manufactured more than one (1) year to the time of production of the fuze assembly, and be conditioned at - 50 F for a minimum of 24 hours. The sample shall be assembled to a suitable fixture (see fig. 4) and tested within 30 seconds after removal from the conditioning chamber and a two (2) oz max, steel ball shall be dropped at a height of eight (8) inches max. onto the top of the firing pin. The fixture shall be capable of measuring the time elapsed between the ignitions of the primer to the flash of the pyrotechnic mix. Any item functioning less than 1.8 seconds shall be classed as a critical and this lot of M53 delay elements shall not be used for production. Any item that does not function or function above 6.0 seconds shall be classed defective. Two (2) or more defectives shall constitute failure of the deterioration test and the lot shall not be used in fuze production without written PCO direction."

4.5.23 Add paragraph as follows:

"4.5.23 Test for deterioration of M76 stab detonator. One hundred (100) samples shall be randomly selected from a lot of M76 stab detonators, which were manufactured more than two years to the time of production of the fuze assembly, and be conditioned at – 50 F for a minimum of 24 hours. The samples shall be assembled to a suitable fixture and tested within 30 seconds after removal from the conditioning chamber and a two (2) oz., maximum, steel ball shall be dropped at a height of eight (8) in., maximum, onto the top of a firing pin. The fixture shall be capable of measuring the time elapsed between the ball striking the firing pin to the functioning of the detonator. Five (5) samples with elapsed time less than 30 milliseconds or greater than 70 milliseconds, or two (2) or more non-functioning detonators or output less than 0.567 inch diameter, minimum, perforation in a 0.131 inch thick, minimum, lead disc with a 1/8 inch hole, maximum, in its center shall constitute failure of the deterioration test and lot shall not be used in fuze production without written PCO direction."

4.5.24 Add new paragraph as follows:

" 4.5.24 Test for deterioration of M98 stab detonator. One hundred (100) samples shall be randomly selected from a lot of M98 stab detonators, which were manufactured more than two (2) years to the time of production of the fuze assembly, and be conditioned at – 50 F for a minimum of 24 hours. The samples shall be assembled to a suitable fixture and tested within 30 seconds after removal from the conditioning chamber and a two (2) oz., maximum, steel ball shall be dropped at a height of three (3) in., maximum, onto the top of a firing pin. Two (2) or more non-functioning detonators or detonator output less than 7/32 inch diameter, minimum, perforation in a 0.131 inch thick, minimum, lead disc shall constitute failure of the deterioration test and lot shall not be used in fuze production without written PCO direction."

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4.5.25, Add paragraph as follows:

"4.5.25 Lead assembly output test. The test shall be performed on the lead assembly (dwg. 9234683). A fixture shall be used to hold the assembly in central contact with the lead disk (dwg. 8816368). A ball (dwg. 8816014) shall be dropped a distance of five (5) inches □ 0.1 inch onto a firing pin (dwg. 9200012). The distance is measured from the bottom of the ball to the top of firing pin. The firing pin shall strike and initiate an M98 Detonator (dwg. 9243921) which in turn will function the lead assembly. Obervation shall be made for compliance with requirements as defined in paragraph 3.20. Other suggested, but not mandatory, equipment which could be utilized in conjunction with the setup are: holder (dwg. 8816367), bushing (dwg. 8827829), and base (dwg. 8802085). Government approval, of the test fixture to be used, is required."

PAGE 103

6.5 Delete in its entirety and substitute:

"6.5 Test for deterioration of detonators and delay elements. If the total elapsed time between the manufacuter of the M53 delay element and the production of the fuze assembly lot exceed one (1) year, or the manufacturer of the M76 and M98 detonators and the production of the fuze assembly lots exceed two (2) years, or if the delay elements and ddetonators have been subjected to adverse conditions, however brief, then testing will be performed on the delay elements and detonators to verify functionality no more than two weeks prior to the assembly into fuzes. For the M53 dleay element, testing will be in accordance with 4.5.22. For the M76 stab detonator, testing will be in accordance with 4.5.23. For the M98 stab detonator, testing will be in accordance with 4.5.24. If the testing indicate that the delay element or detonator lots did not meet their respective requirements, then the lot will not be used for production without authorization of the PCO.

This test will be conducted at Government expense without cost to the detonator and delay element supplier, or the fuze manufacturer, and will not consitutute a basis for rejection against either one, except where deterioration had occurred as a direct result of carelessness in handling, storage, etc., while the detonators and delay elements under the jurisdiction of either contractor."

PAGE 104

6.7.2, Under quantity column, delete "15" and substitute "5."

PAGE 105

6.9, Add new paragraph as follows:

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"6.9 Special defect. A defect, other than critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

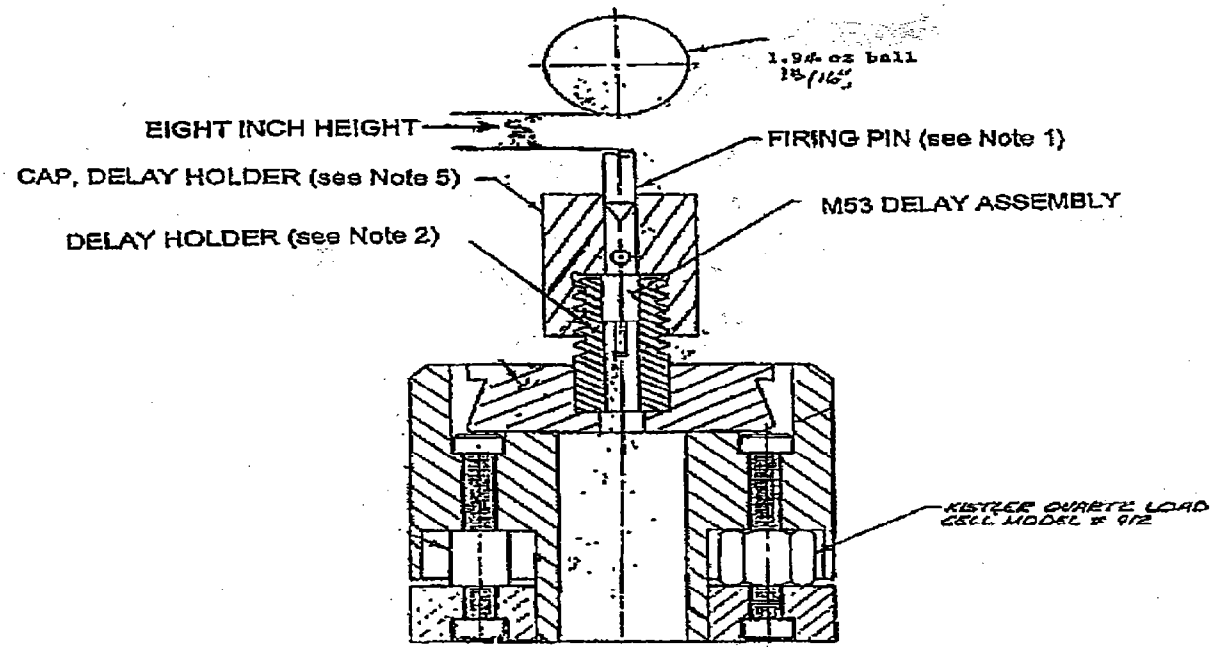
a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item."

PAGE 109

Add fig. 4 as included page 109a.

M53 Delay Assembly Test Fixture (Outline)



NOTES:

1. The firing pin tip shall have the shape defined on the M935 striker drawing 9246230. The firing pin shall be used only once.
2. The M53 Delay Assembly shall be installed in a holder before temperature conditioning. The cap for the holder shall have sufficient length to guide the firing pin over the M64 Primer.
3. The steel ball 1.96 ounce (15/16" Diameter) shall fall freely to impact the firing pin.
4. The start and stop of the delay time can either be acoustic, light detecting or shock wave. It maybe a combination of two types. The importance is the consistency of the responses to the M53 stimuli.
5. The M53 Delay Holder cap shall have four radial vents (~0.080" dia.) just above the M53 to adequately vent the F33B products.

FIGURE 4

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